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EXPERIMENT – 14

Study and Demonstration of Brute Force Password Attack in a Controlled Kali Linux Environment

Aim

Tostudyand demonstratetheconceptofbruteforcepassword attacksinacontrolledKali Linux environment to understand password vulnerabilities and the importance of strong authentication mechanisms.

Software Requirements

HostOperatingSystem:	Windows / Linux / macOS
VirtualizationSoftware:	Oracle VirtualBox
GuestOperatingSystem:	Kali Linux
StudyTools:	Open-source password auditing utilities
SystemRequirements:	

- RAM: Minimum 4 GB (8 GB recommended)
- Disk Space: Minimum 30 GB
- Processor: Intel/AMD with Virtualization support

Introduction

A brute force password attack is a method where all possible password combinations are systematically tried until the correct one is found.
Such attacks exploit weak passwords, poor authentication policies, and lack of account lockout mechanisms.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. OpenOracleVirtualBox.
2. StartKaliLinux.
3. LoginusingKalicredentials.

Step 2: Set Up a Controlled Test Environment

1.Use a locally created test account or dummy authentication service.
2. Ensure no real systems or credentials are involved.
3. Confirm the environment is isolated from external networks.

Step 3: Study Brute Force Attack Workflow

1. Understand how login mechanisms validate credentials.
2. Analyze how attackers automate repeated login attempts.

Step 4: Educational Demonstration (Simulation Only)

1. Simulate repeated authentication attempts using dummydata.
2. Observe system responses such as delays or access denial.
3. Record observations without executing real attacks.

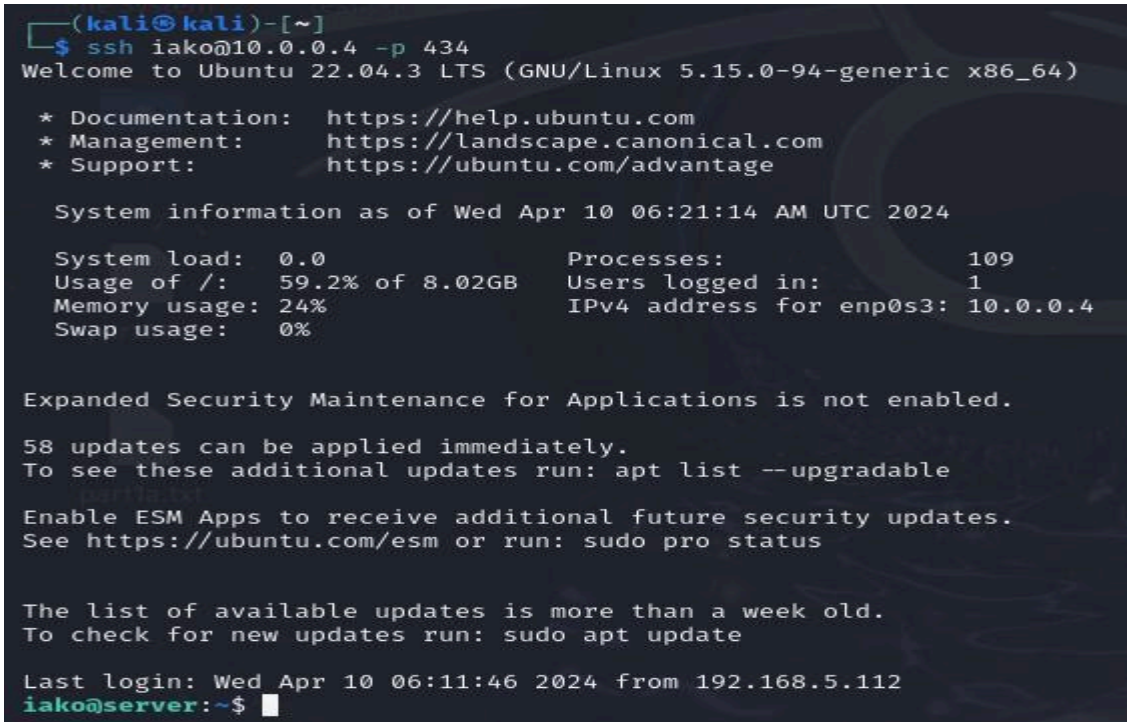
Step 5: Analyze Security Weaknesses

- Weak or short passwords
- No rate limiting
- Missing account lockout policies
- Lack of multi-factor authentication

Step 6: Study Prevention and Mitigation Techniques

- Strong password policies
- Account lockout and rate limiting
- CAPTCHA mechanisms

Output



```
┌──(kali㉿kali)-[~]
└─$ ssh iako@10.0.0.4 -p 434
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-94-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Wed Apr 10 06:21:14 AM UTC 2024

  System load:  0.0                Processes:               109
  Usage of /:   59.2% of 8.02GB    Users logged in:         1
  Memory usage: 24%                IPv4 address for enp0s3: 10.0.0.4
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

58 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Wed Apr 10 06:11:46 2024 from 192.168.5.112
iako@server:~$
```

Result

The study and demonstration of brute force password attacks were successfully completed in a controlled Kali Linux environment.

The experiment emphasized the importance of strong passwords and robust authentication mechanisms to prevent unauthorized access.